Computer Architecture Study

Master:

Dr. Safari

Project:

Single-Cycle Micro-Arch Design for RISC-V ISA

Contributors:

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Project Workflow

1. **RTL Elements Identification**

   Begin by identifying and designing the necessary RTL components for the project, including modules such as **Adder**, **ALU**, and other basic building blocks.

2. **Data Path**

   Next, construct the **Data Path** by interconnecting the RTL elements to enable proper data flow and processing within the system.

3. **Controller**

   Design the **Controller** responsible for generating control signals that coordinate the operations of the Data Path and RTL components.

4. **Assembly Code Implementation for Testing**

   Finally, write and execute an **assembly program** to verify the functionality and correctness of the entire design.

RTL Elements

The Verilog codes for the required RTL elements are available at: `”$PATH/TO/PROJECT/”code/elements/`

For this part, no testbench or specific design is provided, as it is not necessary.

But the function table of the ALU, and Imm_Ext is available at: `”$PATH/TO/PROJECT/”tables/`

ALU: It has two **32Bit A**, and **B** inputs with a **3Bit f** select that puts **A op(f) B** on **32Bit Y** output; Also there are two signal outputs: **zer**, active when **Y = 0** and and **lt**, active when **op(f) = slt && A < B**, don’t care when **op(f) != slt**

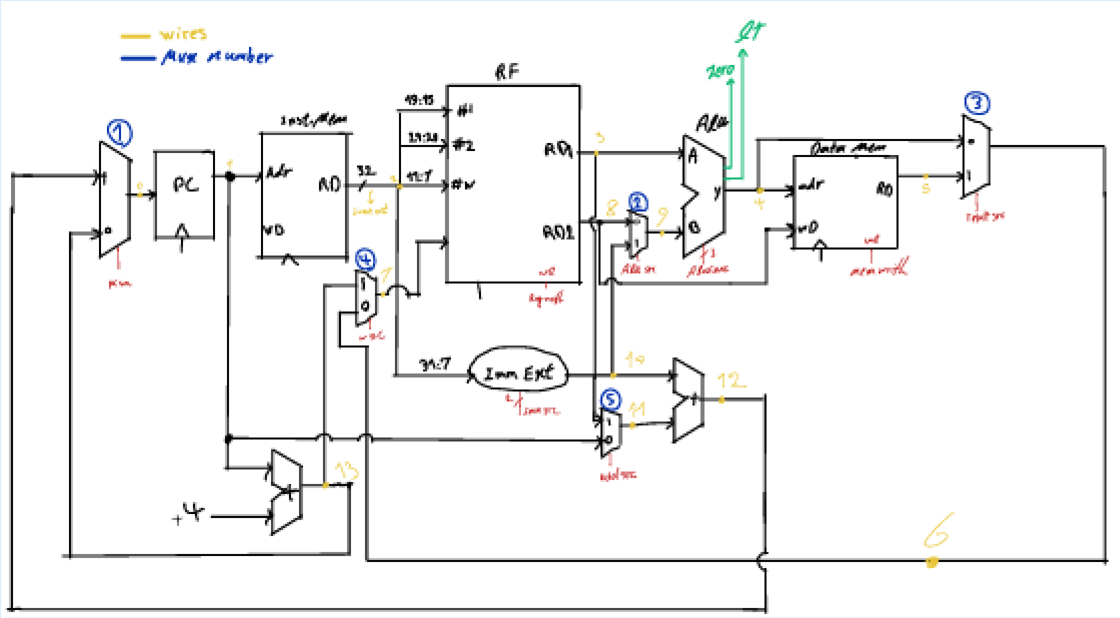
Other elements’ functionality is simple and obvious.

Data Path design

Design:

DP (Data Path) is at first designed on the paper, it is available at: ”$PATH/TO/PROJECT/”papers/

Also you can see down here:



Data Path schematic design

Wires’ and MUXs’ Indexes are entirely as same as the picture above, also Wire[6] is named InstOut, which is noticeable.

Code:

After design, the schematic is implemented by Verilog HDL, which is available at: ”$PATH/TO/PROJECT/”code/

Also a test bench is generated to test its' functionality, available at: "$PATH/TO/PROJECT/"code/test/

we expected that the hardware execute a simple **DPTest** assembly code available at: "$PATH/TO/PROJECT/"assembly/

Simulation:

At the end we run the simulation on "**ModelSim**" and shots of the result is available down here:

Controller:

Put notes for Controller here

Assembly:

Put notes for Assembly here
